

Spreading 47 Points in the Plane

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Abstract

We place 47 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 44.59452+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 48-point configuration gives 44.98169; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 47 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 44.59452\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 47$ is not on the page. Deleting a point from our 48-point configuration gives 44.98169; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n47.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
2623207365252310685520745
denominator:
58823529411656514357497
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

47 points, x and y per line. File: friedman-maxmin2d/coords/n47.txt.

39.194721676373	53.277922087035	32.872786406815	53.215670398714
34.809347325386	57.436907611052	38.487012617333	56.831605860860
37.758022774275	57.516130373729	36.402645594400	53.941330770444
33.611611057614	54.294539145044	39.263907954304	54.834552960510
37.220338339613	53.365675861296	34.294831963158	51.806416316521
38.200719288136	53.168563721198	35.419928377913	53.395354158488
34.469503841238	56.007265737887	33.490702736217	52.400870844958
34.756641347458	55.049376364044	37.235050809181	56.663780490332
33.979466372737	56.878967108463	36.777641825752	57.713242513827
35.299384793888	56.565206240476	38.792427605185	52.362411584480
34.670635110399	52.733115833368	37.622632410801	54.281186363850
38.967262330987	55.926547574663	33.866505883458	53.327570361805
38.603013359323	54.084074223752	36.254669860659	56.860892630431
36.312960191871	52.945360626026	33.783516062585	55.279652676199
32.617891580971	54.182639181952	35.764632392157	57.732594001007
33.293478594084	56.151354046776	38.283527005781	55.031665100608
38.109826672876	51.631620255023	37.465200106799	52.396117861499
37.202448525134	51.211305019752	36.557821959057	51.975802626229
35.601128500745	55.608186412000	37.303146057259	55.228777240706
36.574156214201	55.913301753575	35.562086079683	52.279995032437
36.143959951634	51.065462997951	32.789796585942	55.167752713108
35.509613780442	54.391324302907	37.964040652239	55.979255977464
35.186276445518	51.353286384968	36.345835518072	54.939715773951
34.576478001487	54.031800147898		